Unit: 6 Unit 5: Linear Equations
Topic: Scatter Plot and The Best Fit Line

Name: ______________
Date: __________

Advanced (Carolina Reaper)

Q1. What type of correlation is shown by the scatter plot where as x increases, y also increases?
(A) Positive
(B) Negative
(C) None
(D) Inverse
(E) Direct

Q2. A bakery sells x loaves of bread per day, and the profit is y dollars. If the scatter plot shows a strong positive correlation, what can be said about the relationship between x and y?
(A) As x increases, y decreases
(B) As x increases, y increases
(C) As x decreases, y increases
(D) There is no relationship
(E) The relationship is inverse

Q3. The line of best fit for a set of data has the equation $y = 2x + 1$. What is the y-intercept?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Q4. A student's grades are plotted on a scatter plot, with hours studied on the x-axis and grade percentage on the y-axis. If the line of best fit has a slope of 5, what can be said about the relationship?
(A) For every hour studied, the grade decreases by 5
(B) For every hour studied, the grade increases by 5
(C) For every hour studied, the grade remains the same
(D) There is no relationship
(E) The relationship is inverse

Q5. A scatter plot shows the relationship between the number of minutes spent exercising and the number of calories burned. If the line of best fit has the equation $y = 0.5x - 2$, what is the predicted number of calories burned if a person exercises for 30 minutes?
(A) 10
(B) 12
(C) 13
(D) 14
(E) 16

Q6. The correlation coefficient for a set of data is 0.8. What can be said about the relationship between the variables?
(A) Strong positive correlation
(B) Weak positive correlation
(C) Strong negative correlation
(D) Weak negative correlation
(E) No correlation

Q7. A scatter plot shows the relationship between the amount of money spent on advertising and the number of products sold. If the line of best fit has a slope of 10, what can be said about the relationship?
(A) For every dollar spent on advertising, 10 more products are sold
(B) For every dollar spent on advertising, 10 fewer products are sold
(C) For every dollar spent on advertising, the number of products sold remains the same
(D) There is no relationship
(E) The relationship is inverse

Q8. The equation of the line of best fit for a set of data is $y = -3x + 2$. What is the predicted value of y when $x = 4$?
(A) -6
(B) -4
(C) 2
(D) 6
(E) 10

Open Ended Questions (FRQ)

Q9. A company's daily profit is plotted against the number of items sold. The scatter plot shows a strong positive correlation. If the line of best fit has the equation $y = 2x + 5$, and the company sells 15 items, what is the predicted daily profit? Show your work and explain your answer.

Q10. A student's grades are plotted against the number of hours studied. The scatter plot shows a correlation coefficient of 0.9. If the line of best fit has the equation $y = 3x - 2$, and the student studies for 6 hours, what is the predicted grade? Explain how you would use this information to advise the student on how to improve their grade.

Chef's Tips

- Tip 1: Emphasize the importance of accurate calculations and clear explanations.
- Tip 2: Encourage students to use real-world examples to illustrate their understanding of the concepts.
- Tip 3: Remind students to check their work and ensure that their answers are reasonable and well-supported.